

NMEC595 - Research Methodology

Lecture 1: Research Methodology

Prof. Suparno Bhattacharyya

suparno@iitism.ac.in

224A, Mechanical Engg. Building

Winter semester 2026

Dept. of Mechanical Engineering, IIT(ISM) Dhanbad

Table of Contents

- 1 Understanding Research
- 2 Objectives & Motivation
- 3 Types of Research
- 4 Research Approaches
- 5 Significance & Applications
- 6 Methods & Methodology
- 7 Scientific Method
- 8 Research Process

What is Research?

- **Definition:** A systematic, scientific search for pertinent information on a specific topic
- **Nature:** An original contribution to existing knowledge through objective and systematic methods
- **Core Elements:**
 - Search for knowledge (inquisitiveness)
 - Systematic investigation
 - Scientific approach
 - Voyage of discovery
- **Purpose:** To discover answers to questions through scientific procedures

Research as a Process

According to Clifford Woody, research comprises:

- 1 Defining and redefining problems
- 2 Formulating hypotheses or suggested solutions
- 3 Collecting, organising and evaluating data
- 4 Making deductions and reaching conclusions
- 5 Testing conclusions to determine fit with hypothesis

In essence: **Problem** → **Hypothesis** → **Data** → **Analysis** → **Conclusions**

Objectives of Research

Research objectives fall into four broad categories:

- ① **Exploratory**: Gain familiarity or achieve new insights into a phenomenon
- ② **Descriptive**: Portray accurately the characteristics of individuals, situations or groups
- ③ **Diagnostic**: Determine frequency of occurrence or association with other phenomena
- ④ **Hypothesis-Testing**: Test causal relationships between variables

Motivation for Research

Why do people undertake research?

- Desire for research degree
- Challenge of solving unsolved problems
- Intellectual joy of creative work
- Service to society
- Respectability and recognition
- Curiosity about new phenomena

Classification I: Descriptive vs. Analytical

Descriptive Research

- Fact-finding enquiries; surveys
- Describes state of affairs as it exists
- Researcher has no control over variables
- Examples: shopping frequency, consumer preferences

Analytical Research

- Uses facts or information already available
- Makes critical evaluation of material
- Goes beyond description to analysis

Classification II: Applied vs. Fundamental

Applied Research

- Solves immediate problems
- Addresses social or business needs
- Examples: market research, copy research, evaluation research
- Focused on practical outcomes

Fundamental Research

- Knowledge for its own sake
- Formulates theory and generalizations
- Examples: natural phenomena, pure mathematics
- Broad base of applications

Classification III: Quantitative vs. Qualitative

Quantitative Research

- Based on measurement of quantity or amount
- Expresses phenomena in numerical terms
- Statistical analysis and rigorous methods

Qualitative Research

- Concerned with quality or kind
- Investigates reasons for human behavior
- Uses in-depth interviews, projective techniques
- Important in behavioral sciences

Classification IV: Conceptual vs. Empirical

Conceptual Research

- Related to abstract ideas or theory
- Develops new concepts
- Reinterprets existing concepts
- Used by philosophers and thinkers

Empirical Research

- Relies on experience or observation
- Data-based with experimental design
- Conclusions verified by observation
- Most powerful support for hypotheses

MCQ 1: Identify the Research Type

Research activity:

A faculty member analyzes five years of published crash-test datasets and government reports to evaluate whether a proposed change in seat-belt pretensioner settings reduces chest injury metrics. No new experiments or surveys are conducted; the work focuses on interpreting existing information and drawing conclusions.

Question: This activity is best classified as:

- ☐ A. Descriptive research
- ☐ B. Analytical research
- ☐ C. Qualitative research
- ☐ D. Conceptual research

MCQ 2: Identify the Research Type

Research activity:

A research group conducts semi-structured interviews with first-year engineering students to understand why they skip tutorial sessions. The team codes the interview transcripts to extract recurring themes (e.g., time constraints, classroom climate, perceived usefulness) rather than computing numerical summaries.

Question: This activity is best classified as:

- ☐ A. Quantitative research
- ☐ B. Qualitative research
- ☐ C. Empirical research
- ☐ D. Fundamental research

Other Classification Dimensions

- **By Time:** One-time research vs. Longitudinal research
- **By Setting:** Field research vs. Laboratory research vs. Simulation research
- **By Approach:** Clinical/Diagnostic vs. Exploratory vs. Formalized
- **By Sources:** Historical research using documents and records
- **By Decision:** Conclusion-oriented vs. Decision-oriented research
- **Specialized:** Operations research for quantitative decision-making

Quantitative Approach

- Generates data in quantitative form
- Subjected to rigorous quantitative analysis

Subtypes:

- Inferential: Form database from sample to infer population characteristics
- Experimental: Control variables; manipulate to observe effects
- Simulation: Construct artificial environment to generate controlled data

Qualitative Approach

- Subjective assessment of attitudes, opinions, behavior
- Result is function of researcher's insights and impressions
- Non-quantitative or not subjected to rigorous statistical analysis
- **Common Techniques:**
 - Focus group interviews
 - Projective techniques (word association, sentence completion)
 - Depth interviews

Significance of Research

"All progress is born of inquiry. Doubt is often better than over-confidence, for it leads to inquiry, and inquiry leads to invention."

Research Benefits:

- Inculcates scientific and inductive thinking
- Promotes logical habits of thinking
- Increases amounts of knowledge for progress
- Provides basis for policy-making

Research in Governance & Policy

Government relies on research for:

- Budget allocation and resource distribution
- Equating needs to available revenues
- Devising alternative policies and examining consequences
- Addressing all facets of national existence
- Collecting economic and social structure information

Three Phases of Research in Policy:

- 1 Investigation of economic structure through fact compilation
- 2 Diagnosis and analysis of events and underlying forces
- 3 Prognosis: prediction of future developments

Research in Business & Industry

Critical research areas:

- **Market Research:** Structure and development of market; efficient policies
- **Operations Research:** Mathematical and analytical techniques for optimization
- **Motivational Research:** Why people behave as they do; consumer behavior
- **Business Budgeting:** Based on sales forecasts and market estimates

Result: Replace intuitive decisions with logical and scientific decisions

Research in Social Sciences

- Studies social relationships and solves social problems
- Provides intellectual satisfaction of knowing
- Contributes to practical concerns and human relations
- **Dual Emphasis:**
 - Develop principles for understanding and predicting human interactions
 - Provide practical guidance for solving immediate human relations problems
- Benefits to researchers and practitioners

Research Methods vs. Research Methodology

Research Methods

All techniques/methods used in conducting research; behavioral and instrumental approaches for:

- Collecting data
- Establishing relationships
- Evaluating accuracy

Research Methodology

Systematic way to solve research problems; includes:

- Logic behind methods chosen
- Justification for specific techniques
- Understanding assumptions
- Criteria for applicability

Why Methodology Matters

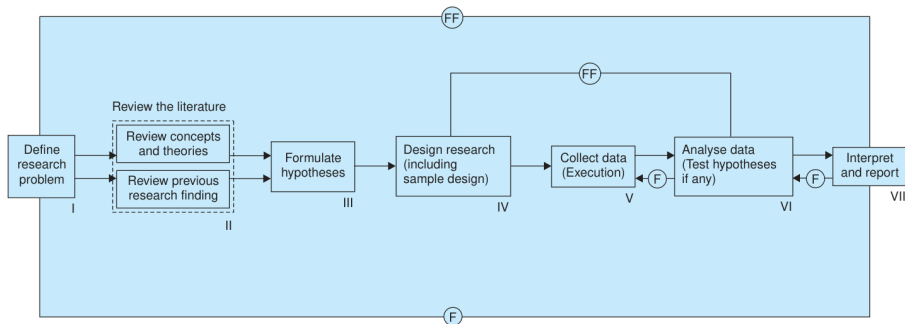
- **Broader Scope:** Wider than just methods
- **Transparency:** Specify decisions clearly so others can evaluate
- **Accountability:** Expose decisions to evaluation before implementation
- **Replicability:** Enable verification through reproduction

Example: Architect consciously evaluates why doors, windows, materials are chosen for a building—similarly, researcher must justify all research decisions.

Postulates of Scientific Approach Adopted During Research

- 1 Evidence-based (empirical)
- 2 Utilizes relevant concepts
- 3 Committed to objective considerations only
- 4 Presupposes ethical neutrality (adequate and correct statements)
- 5 Results in probabilistic predictions
- 6 Methodology made known to all for critical scrutiny and testing via replication
- 7 Aims at formulating most general axioms or scientific theories

Research Process in Flow Chart



Where (F) = feed back (Helps in controlling the sub-system to which it is transmitted)

(FF) = feed forward (Serves the vital function of providing criteria for evaluation)

Formulating Research Problem

Key Aspects:

- Identify specific problem from general area of interest
- Resolve ambiguities relating to problem
- Consider feasibility of solution
- Set up working formulation

Two Essential Steps:

- 1 Understand the problem thoroughly (discuss with colleagues/experts)
- 2 Rephrase into meaningful analytical terms

Critical Importance: Unambiguous problem definition discriminates relevant from irrelevant data.

Why Know How Research is Done?

- **For Researchers:** Essential tools of trade; develop disciplined thinking
- **For Evaluators:** Judge research results with confidence; make informed decisions
- **For All:** New intellectual tool for objective world-viewing
- **For Consumers:** Evaluate research adequacy; take rational decisions

Provides training in:

- Material gathering and organization
- Data collection techniques
- Statistical analysis and interpretation
- Evidence evaluation

Key Takeaways

- Research is systematic, scientific search for knowledge and solutions
- Multiple types serve different purposes (exploratory, descriptive, diagnostic, hypothesis-testing)
- Research approaches vary: quantitative, qualitative, or mixed
- Understanding methodology, not just methods, is crucial
- Scientific method ensures objectivity and replicability
- Research process is iterative with overlapping steps
- Knowledge of research methodology valuable for all professionals